Missingness Bias Calibration in Feature Attribution Explanation

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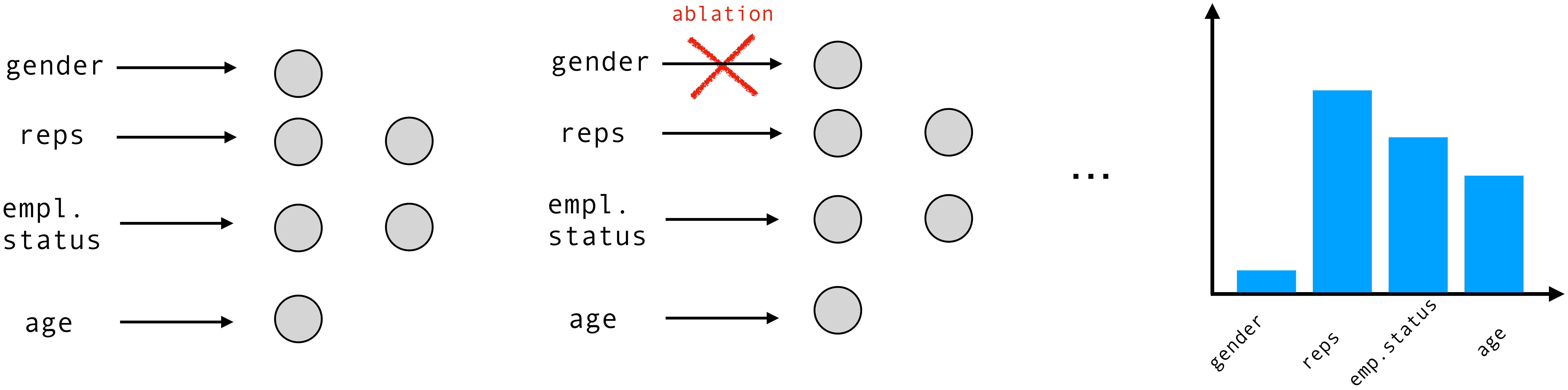
(Edoardo Canti - workshop presentation II - Deep Learning Applications 25/26)

Features Attributions methods

Feature Attribution methods: techniques to evaluate features importances to the model output.

Iteratively remove features and evaluate model changes in output to assess each feature importance (LIME/SHAP)

Of course this is not possible:



Approximating Ablation & Missingness bias

Ablation can be approximated via input pre-processing.

`a gentle, compassionate drama about grief and healing.`

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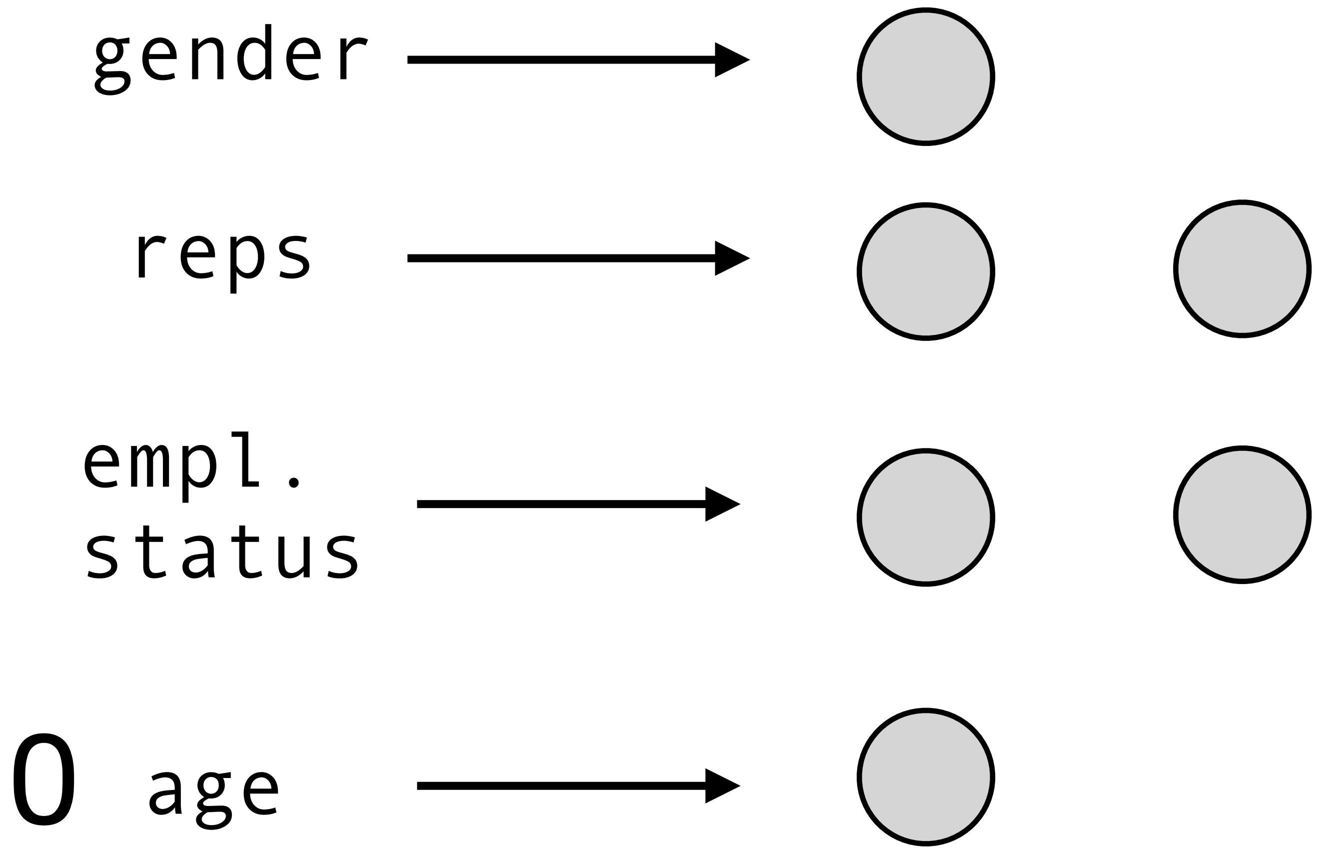
eg2: Masking a natural language input(*)

eg1: Masking a subset of pixels (taken from the paper)



eg3: substituting 0 to age feature

Out Of Distribution inputs corrupts model behavior systematically: model is watching to unrealistic data, on which it wasn't trained!

$\Downarrow$

Predictions uncontrolled shift: **Missingness Bias!**

On the Missingness Bias

1) Accuracy degradation with respect to a particular class (fig1)

2) The model becomes untrustworthy: if predictions on ablated inputs are not correct, the model is giving importance to not relevant features (fig2)

3) Quantification:

$$\mathrm{MB}(f) = D_{\mathrm{KL}}\big(\mathrm{Class}(f(X')) \parallel \mathrm{Class}(f(X))\big)$$

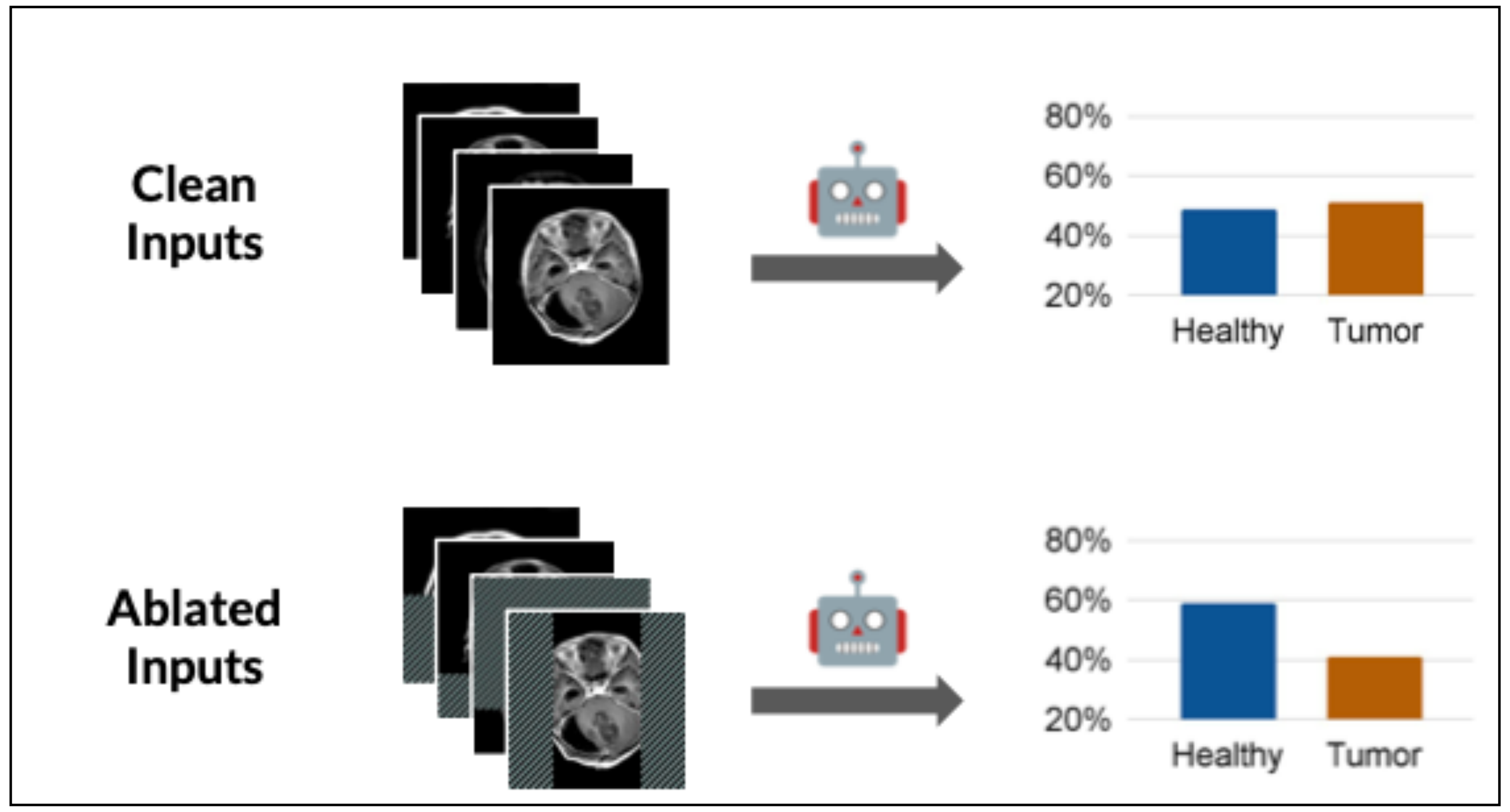


fig1: low accuracy on “Healthy” (tumor is visible in ablated input) (from the paper)

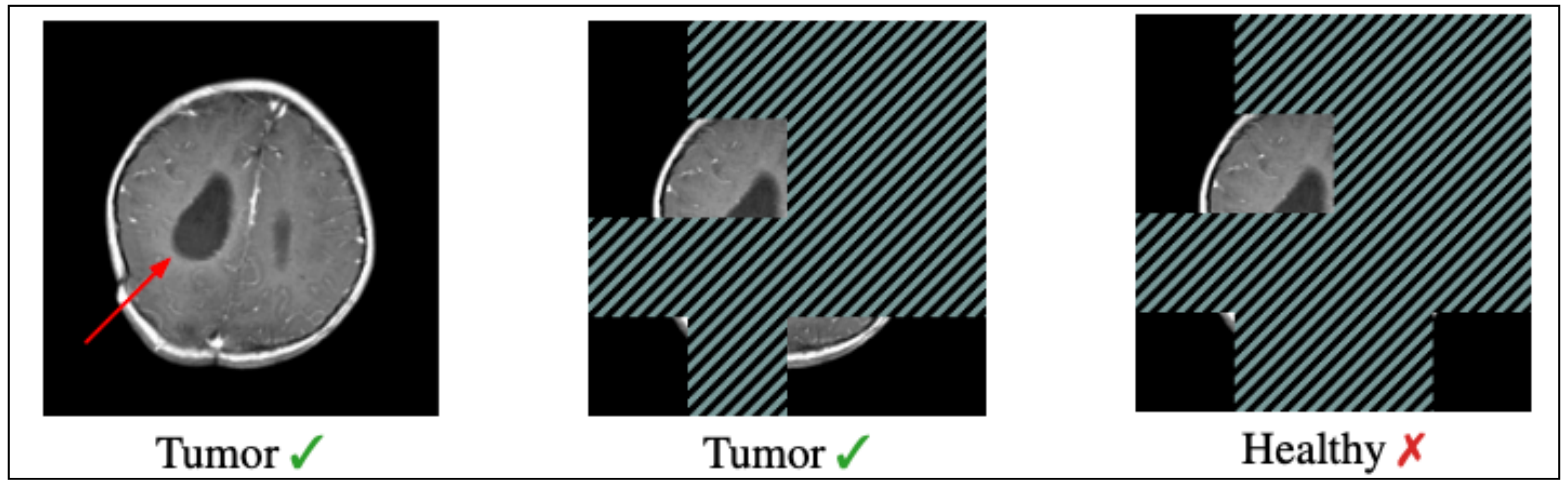


fig2: Tumor is visible, masking pixels without tumor (leaving tumor visible!) predict the wrong class! (from the paper)

Mitigation methods proposed in literature

1) Replacement: try to substitute feature values with more complex methodologies (instead of zero, mean etc…). In Images authors consider in-paint methods for creating realistic images patches using generative models (complex approach, need to be validated)

2) Training Based: use and augmentation modules (like in Supervised Contrastive Learning) with the only goal to mask inputs. The model will get “comfortable” with missing data (expansive and are you sure to have access to the model?)

3) Architectural Based: modify the model itself to consider irrelevant the special tokens (is insanely unscalable!)

Proposal: MCal

Given the **classifier** $f_\theta : \mathbb{R}^n \to \mathbb{R}^m$ process the input and get **its logits** $z = f_\theta(x)$

Consider a **linear Calibrator** $R_\theta : \mathbb{R}^m \to \mathbb{R}^m$ that elaborates those logits: $R_\theta(z) = Wz + b$

The Calibrator is fitted by using a CE-Loss between its output (that are based on the ablated input) and the class predicted by the **classifier** on the not ablated output.

f(x) —> logits = [5, 1, 2]

$$\mathscr{L} = \mathrm{CrossEntropy}\left(R\left(f(\tilde{x})\right), \mathrm{OHE}\left(\mathrm{Class}(f(x))\right)\right)$$

x’ <— ablation(x)

f(x’) —> logits = [2, 13, 7]

x’ involved features must be very important!

SHAP

Authors now suggest that this new calibrated model can be used to replace f

R (f(x’), f(x)) = new_logits

Results

	Dataset	Base	Replace	Retrain	Arch	TempCal	PlattCal	MCal (✓)
Vision	Brain MRI	1.18 e−1	1.51 e−1	**6.70 e−4**	1.40 e−1	1.16 e−1	1.27 e−1	7.43 e−3
	CheXpert	1.70 e−1	9.70 e−2	2.67 e−2	1.50 e−1	1.65 e−1	2.02 e−1	**8.82 e−3**
	BreakHis	1.87 e−1	4.20 e−1	2.19 e−2	1.54 e−1	1.86 e−1	1.66 e−1	**4.29 e−3**
Language	MedQA	1.61 e−1	1.50 e−1	1.70 e−1	2.68 e−2	1.57 e−1	9.48 e−2	**9.44 e−4**
	MedMCQA	1.89 e−1	2.59 e−1	1.52 e−1	1.40 e−1	7.81 e−1	1.13 e−1	**9.01 e−3**
Tabular	PhysioNet	1.17 e−1	1.20 e−1	5.59 e−3	8.14 e−2	1.17 e−1	1.19 e−1	**5.01 e−3**
	Breast Cancer	1.02 e−1	1.44 e−1	5.68 e−3	2.13 e−1	1.02 e−1	1.08 e−1	**1.92 e−5**
	CTG	1.06 e−1	7.02 e−2	6.61 e−3	2.85 e−1	1.06 e−1	9.20 e−2	**3.35 e−3**

Tab1: evaluating missingness bias via KL-Divergence on different mitigation strategies (from the paper)

Tab1 shows that on different MCal outperforms other mitigation methods on different types of data.

The only exception comes for Brain MRI, in which MCal is outperformed by a Retrain pipeline (of course we should consider that this implies to retrain the full model)

Conclusions

With respect to existing methods for mitigating missingness bias, that are expensive (full retrain), not completely reliable and bias-affected (generative models) and not scalable (not all models are open!); MCal helps mitigating this bias by just adding a simple Linear Classifier that is trained in order to re-allign outcomes of the ablated input with, "more plausible" predictions.

Personal doubt: if the Calibrator would be able to reproduce the exact same logits produced by the Classifier, we would have a zero difference in the logits predicted over the ablation dataset and the "correct" dataset (SHAP would says that there is no feature importance).

To which extent can we assume that MCal doesn't overfits and produces exact same logits?

Thanks!

[1] Sridhar, Shailesh, Anton Xue, and Eric Wong. "Missingness Bias Calibration in Feature Attribution Explanations." *International Conference on Learning Representations*. Vol. 2026. 2026.